Instructor Run Sheet

EBIKE101 | CTS Cohort 6 | Day 33 | Monday, August 31, 2026

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EBIKE101 Day 1: Introduction, Rider Safety, and System Inspection

Safe micro-mobility starts before the throttle moves.

Official focus

Students begin the Electric Micro Mobility block by treating an e-bike as a vehicle system, not a toy. Day 1 introduces rider safety, motor/throttle control boundaries, the difference between conventional bicycle systems and electric assist systems, and the pre-ride inspection standard students will use for the full two-week section.

The nearly completed EDM101 subwoofer may be shown as a class culture carryover only. It is not part of the official EBIKE101 curriculum, grading, or learning targets.

Today's big question

What makes an e-bike different from a bicycle, and what must be proven safe before anybody rides it?

Core rule

No ride, no throttle test, and no powered wheel spin until the bike passes a visible safety check.

Learning targets

Students will be able to:

1. Identify the main systems on a conventional bicycle: frame, fork, wheels, tires, brakes, drivetrain, cockpit, and contact points.

2. Identify the main e-bike systems: battery, BMS concept, motor, controller, display, throttle or pedal assist sensor, wiring, connectors, charger, and cutoffs.

3. Explain why rider control, braking, tire condition, wiring security, and battery condition must be checked before operation.

4. Complete a pre-ride inspection checklist and record pass/fail notes.

5. Trace the basic e-bike flow: rider input -> controller decision -> battery power -> motor output -> wheel motion.

Readiness and materials

Before students arrive

- Stage one e-bike or conversion bike where students can walk around it.
- Stage one conventional bike if available for comparison.
- Keep battery disconnected or system powered off until the safety boundary is taught.
- Mark a no-ride/no-throttle area around the bike.

- Print student packets and quiz sheets.
- Put inspection categories on the board:

- mechanical

- electrical

- rider control

- route/environment

- documentation

Materials

- E-bike or conversion bike
- Conventional bike, if available
- Bike stand, if available
- Helmets
- Gloves and eye protection as appropriate
- Tire pump and pressure gauge
- Basic bike tools: hex keys, wrench set, tire levers
- Multimeter for demonstration only
- Cones or tape for controlled area planning
- Zip ties or Velcro straps for loose-wire examples
- Printed student packets, quiz, answer key, and lab sheets

Five-hour run of show

Time	Block	Instructor move
10:00-10:10	Morning landing	Attendance, settle, phones down, quick context: EBIKE101 starts today. Ask: "What can go wrong if a powered bike is treated like a normal bike?"
10:10-10:30	Morning quiz	20-question readiness quiz. Grade for safety thinking and vocabulary, not punishment. Fix unsafe assumptions immediately.
10:30-10:50	Course launch	Explain the two-week arc: inspect -> understand -> plan -> convert -> wire -> troubleshoot -> commission -> certify.
10:50-11:15	Bicycle baseline	Walk around a normal bike. Identify frame, fork, wheels, tires, brakes, chain/drivetrain, handlebars, seat, and contact points.
11:15-11:45	E-bike added systems	Add battery, controller, motor, display, throttle/PAS, wiring, charger, cutoffs, and connectors. Keep the bike powered off.
11:45-12:00	Safety boundary	Teach no ride/no throttle/no spin until the pre-ride checklist passes. Discuss stop-work authority.
12:00-12:30	Lunch/reset	Keep bikes staged. Reset tools and packets.

12:30-1:10	Guided inspection demo	Instructor models the pre-ride inspection: tires, brakes, steering, axle/wheel security, drivetrain, battery mount, wiring, connectors, controls.
1:10-1:55	Lab A: team inspection	Teams inspect bike systems using the student checklist. They mark pass/fail/needs instructor review.
1:55-2:10	Break	Reset, hydrate, keep students out of the no-ride area.
2:10-2:40	Lab B: system map	Teams draw the e-bike flow from rider input to motor output and name what could fail at each step.
2:40-2:55	Debrief	Each team reports one mechanical risk, one electrical risk, and one rider-control risk.
2:55-3:00	Closeout	Collect packets. Preview Day 34: conventional bicycle mechanics, because electricity does not fix loose wheels.

Instructor notes

- Do not start with speed. Start with control.
- Do not let "it turns on" become "it is safe." Those are different claims.
- If the bike is incomplete or unsafe, the lab still works as an inspection and planning exercise.
- If students are excited about riding, use that energy but hold the line: inspection first.
- A bike can be mechanically unsafe even if the electrical system is perfect.
- A bike can be electrically unsafe even if it pedals like a normal bike.
- Do not use public-road law details as the lesson unless verified separately. This class is about controlled training use and safe inspection habits.

Stop-work triggers

Pause the lab if any of these happen:

- A student attempts to ride, throttle, or spin the powered wheel without instruction.
- The bike tips, falls, or becomes unstable in the stand.
- Battery damage, swelling, heat, odor, water intrusion, or unknown charger condition is observed.
- Brakes fail a squeeze/roll check.
- Wheel/axle security is uncertain.
- Wiring can contact moving parts, tire, chain, brake rotor, or crank.
- A student jokes about defeating safeties, bypassing cutoffs, or "just sending it."
- A team cannot explain the next action safely.

Completion standard

By the end of Day 1, each student should be able to say:

"Before operating an e-bike, I must verify the mechanical bike, the electrical assist system, the rider controls, and the riding area. Power only comes after control is proven."

Backup plan

If no e-bike is available today:

1. Use a conventional bike as the physical inspection object.

2. Draw or project an e-bike block diagram.

3. Have teams label where the battery, motor, controller, display, throttle/PAS, charger, and cutoffs would fit.

4. Complete the same pre-ride checklist with "not installed" marked where appropriate.

5. Finish with a paper-based hazard analysis.